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Automotive Cyber Security Academy 2023

Chapter 4 – Automotive Hardware Security

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Hardware Trust Anchors

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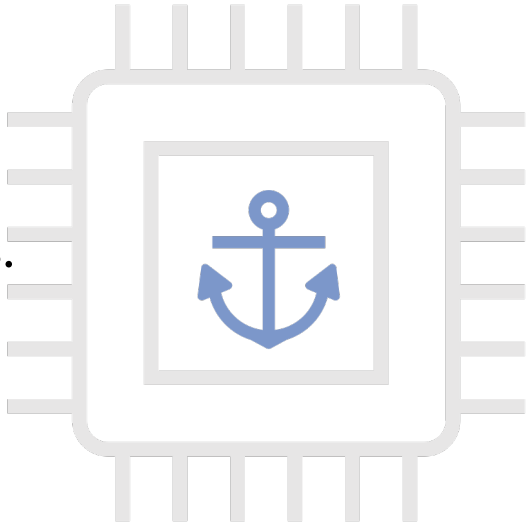

Hardware Trust Anchors (HTAs) are components that provide a dedicated secure area in hardware and foundational security services.

- Often the terms HTA and Hardware Security Module (HSM) are used interchangeably
- No formal definition of the terms HTA or HSM
  - Every supplier defines its own silicon with cryptographic functions as HTA or HSM
  - OEMs (Original Equipment Manufacturers), i.e., car manufacturers, need carefully choose their supplier
- HTAs are basically a set of functions in a Trusted computing base (TCB) of a Hardware device with limited access rights, limited interactions and that is trusted by the system.

Trusted computing base: Totality of protection mechanisms within a computer system, including hardware, firmware, and software, the combination responsible for enforcing a security policy – NIST SP 800-12 Rev 1

HTA Security Services

Secure storage

Crypto generation

Isolated secure environment

Crypto acceleration

Secure Unique Identities

The basis of the main security features implemented in automotive security architectures

HTA Implementations

General examples

- Secure Hardware Extensions (SHE)
- EVITA HSMs and “Bosch HSM”
- Trusted Platform Module (TPM)

- How easy can security services be integrated?
- What crypto building blocks are supported?
- How fast can crypto services run?
- How difficult is for an attacker to retrieve secrets?

Goal: provide secure storage, secure environment for cryptographic operations, crypto-acceleration

HTA Integrations

- HTA integration approaches
  - Integrated into the main system, e.g., Bosch HSM, Firmware TPM
  - External chip connected to the main system via I2C, SPI or other bus system, e.g., TPM chip
  - Secure environments e.g., TrustZone
- Different brands with different (HTA) name
  - Infineon: Aurix HSM / SHE+ driver
  - Renesas: Intelligent Cryptographic Unit (ICU)
  - Freescale: Crypto Service Engine (CSE)
  - ARM: Trust Zone

An HTA is only one component in a security architecture

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Thank You

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